

MINI REVIEW **OPEN ACCESS**

Quantitative and Clinical Pharmacology Perspectives on Next-Generation Malaria Prevention Using Vaccines and Long-Acting Monoclonal Antibodies

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ABSTRACT

Malaria remains a significant global health burden, particularly among pediatric populations in endemic regions. Recent advances in preventive interventions highlight the complementary roles of vaccines and long-acting monoclonal antibodies (mAbs) in reducing infection risk. Vaccines such as RTS,S/AS01 and R21/Matrix-M induce active immunity through polyclonal antibody and CD4⁺ T-cell responses against the circumsporozoite protein (CSP), providing durable, population-level protection. In contrast, mAbs including CIS43LS and L9LS confer immediate, passive immunity, with protection directly linked to systemic exposure and duration above protective thresholds. Translational pharmacokinetic/pharmacodynamic (PK/PD) modeling and physiologically based pharmacokinetic (PBPK) approaches are central to optimizing these interventions. Population PK models, combined with controlled human malaria infection (CHMI) studies, enable model-informed selection of dose, route, and timing to achieve efficacious exposures, while PBPK models facilitate extrapolation to pediatric and special populations by incorporating developmental physiology, protein turnover, and FcRn-mediated recycling. Clinical studies demonstrate robust protection from both vaccines and mAbs, supporting complementary combination strategies. Forward-looking approaches integrate vaccines and long-acting mAbs to leverage broad, durable immunity from vaccines and immediate, predictable protection from antibodies, particularly during seasonal peaks or in high-risk groups. This review emphasizes how quantitative PK/PD and PBPK modeling provide a framework to optimize dosing, predict durability of protection, and guide clinical development across diverse populations. These strategies underscore the value of model-informed translational pharmacology in accelerating development of next-generation malaria preventive interventions against malaria.

1 | Introduction

Malaria remains a major global health challenge, accounting for over 282 million cases worldwide and approximately 610,000 deaths in 80 countries annually [1]. The World Health Organization (WHO) African Region bears a disproportionate share of the global malaria burden, accounting for approximately

95% of all malaria cases and deaths worldwide. In 2024, an estimated 265 million cases and 579,000 deaths occurred in the Region, with children under 5 years of age accounting for nearly three-quarters of malaria-related mortality [1, 2]. Despite significant advances in vector control and chemoprevention, sustained malaria elimination has proven difficult, underscoring the need for effective and durable preventive interventions. In

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this context, vaccines and, more recently, long-acting monoclonal antibodies (mAbs) targeting *Plasmodium falciparum* have emerged as promising tools for malaria prevention.

The first malaria vaccine, RTS,S/AS01, and the second-generation vaccine, R21/Matrix-M, were recommended by the World Health Organization for broader use in 2021 and 2023, marking significant advances in malaria control [3]. Both vaccines target the circumsporozoite protein (CSP) and rely on the induction of antibody- and T-cell-mediated immunity to prevent liver-stage infection [4]. However, their protective efficacy is moderate, variable across populations, and declines over time, particularly in malaria-naïve individuals and immunologically immature children [5, 6]. This variability reflects the mechanistic complexity of vaccine-induced protection, which depends on the magnitude, quality, and durability of the immune response, as well as on prior malaria exposure that can act as immune priming. In addition, the indirect relationship between vaccine dose, immune response, and clinical protection limits the ability to define clear exposure–response (E-R) relationships, complicating quantitative prediction of durability, optimization of dosing schedules, and rational deployment across heterogeneous transmission settings [4, 7].

In contrast, mAbs provide passive immunity by direct administration of high-affinity antibodies against conserved CSP epitopes, offering immediate protection that is independent of host immune status and more directly linked to systemic antibody

concentrations and pharmacokinetics (PK) [8]. These two biological therapeutic modalities, while sharing the common goal of patient protection, engage distinct and complex immunodynamic mechanisms, highlighting the expanding role of pharmacometrics in malaria prevention [9]. This mini-review leverages principles of PK and pharmacodynamics (PD) to compare malaria vaccines and mAbs and highlights how pharmacometrics approaches can support evidence-based decision-making in global malaria control.

2 | Biology of Malaria Infection and Molecular Targets for Preventive Medicine

Figure 1 illustrates the malaria parasite life cycle and the stages targeted by current preventive and therapeutic interventions. Infection begins when an infected *Anopheles* mosquito injects sporozoites into the human host during a blood meal. Sporozoites rapidly migrate to the liver, where they invade hepatocytes and undergo asymptomatic replication during the pre-erythrocytic stage, lasting approximately 5–9 days. Merozoites are then released into the bloodstream, initiating the asexual blood stage by infecting red blood cells, leading to parasite replication, clinical disease, and the formation of gametocytes that enable transmission back to mosquitoes. Because the sporozoite and liver stages involve relatively low parasite numbers and occur before symptom onset, they represent attractive targets for prevention. CSP,

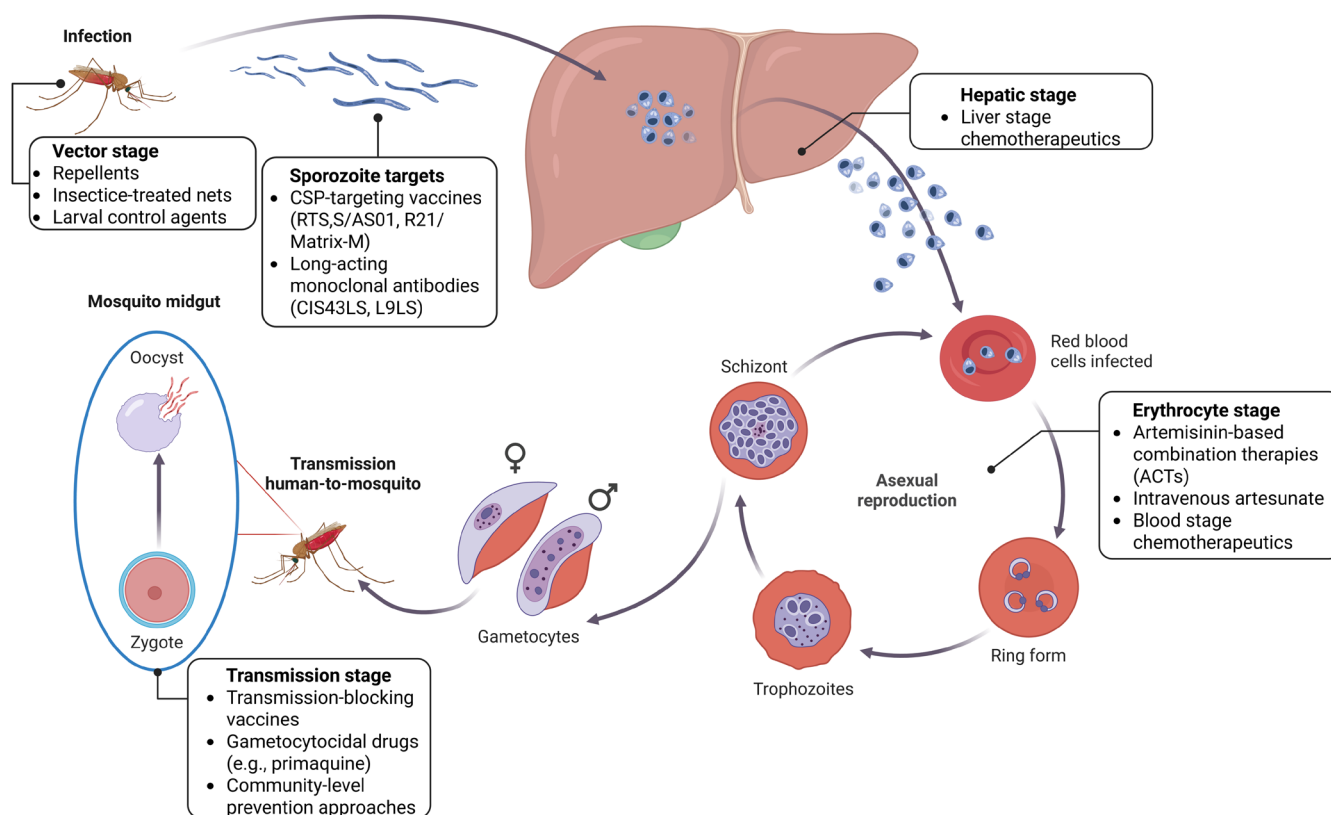


FIGURE 1 | Malaria life cycle and targets for therapeutic and preventive interventions. Schematic of the *Plasmodium* life cycle showing vector, sporozoite, hepatic, erythrocytic, and transmission stages. Key interventions include CSP-targeting vaccines and monoclonal antibodies at the pre-erythrocytic stage, chemotherapeutics at the liver and blood stages, and transmission-blocking strategies.

the major surface antigen on sporozoites, is the primary target of malaria vaccines and monoclonal antibodies. In contrast, antimalarial drugs target blood-stage parasites, while transmission-blocking vaccines prevent parasite development within the mosquito vector.

3 | Pharmacologic Strategies for Malaria Prevention and Treatment

In symptomatic patients, treatment targets the asexual blood stage to rapidly clear parasitemia and prevent progression to severe disease. Artemisinin-based combination therapies (ACTs) are the first-line treatment for uncomplicated malaria, while intravenous artesunate followed by oral ACT is recommended for severe malaria. Chemoprophylaxis with agents such as atovaquone–proguanil, doxycycline, and mefloquine is used for travelers and selected high-risk populations. However, these interventions are administered after infection has occurred or require continuous adherence and remain vulnerable to drug resistance, adverse effects, and repeated exposure in endemic settings. In contrast, preventive interventions act earlier in the parasite life cycle, often before blood-stage infection and clinical disease develop. Vaccines and monoclonal antibodies targeting pre-erythrocytic stages reduce the likelihood of infection establishment, while transmission-blocking vaccines interrupt parasite development within mosquitoes. Collectively, these preventive strategies complement chemotherapeutic treatments and are critical for long-term malaria elimination efforts.

4 | Immune Biologics as Next-Generation Anti-Malarial Therapeutics

Malaria prevention strategies increasingly focus on interventions targeting the pre-erythrocytic stages of *Plasmodium falciparum*, with vaccines and long-acting mAbs representing the two most advanced biologic approaches. Although both primarily target CSP, they differ in their mechanisms of action, PK/PD characteristics, and quantitative modeling approaches.

4.1 | Vaccines—Pharmacokinetics, Pharmacodynamics, Exposure-Response, and Immunogenicity

RTS,S/AS01 and R21/Matrix-M are CSP-based malaria vaccines that prevent infection by inducing anti-CSP antibodies and CD4⁺ T-cell responses before liver invasion (Table 1). Although they have demonstrated population-level benefit, protection remains partial, heterogeneous, and wanes over time. From a PK/PD perspective, vaccine efficacy is driven by the magnitude and durability of immune responses rather than dose alone, resulting in a complex and nonlinear relationship between vaccination and protection. Clinical studies have also reported substantial variability in vaccine-induced anti-CSP antibody concentrations across individuals and dosing schedules, highlighting the importance of quantitative approaches to characterize immune correlates of protection and optimize vaccination strategies.

TABLE 1 | Current and emerging malaria preventive interventions: Vaccines, long-acting monoclonal antibodies, and transmission-blocking candidates.

Class	Candidate	Target/Stage	Development/Use	Key efficacy or activity (endpoint)
Vaccine	RTS,S/AS01	CSP; pre-erythrocytic	WHO recommended (2021)	~36% efficacy against clinical malaria over ~4 years (4-dose, children 5–17 months) [7]; 56% efficacy over 12 months in African children
Vaccine	R21/Matrix-M	CSP; pre-erythrocytic	WHO recommended (2023)	75% efficacy at seasonal sites and 68% at standard sites vs. time to first clinical episode at 12 months [10]
Monoclonal antibody	CIS43LS	CSP junctional epitope; pre-erythrocytic	Phase 2 (completed)	88.2% efficacy (40 mg/kg) and 75.0% (10 mg/kg) vs. time to first infection over 24 weeks [11]
Monoclonal antibody	L9LS	CSP (high-affinity epitope); pre-erythrocytic	Phase 2 (completed)	~77%–88% efficacy against <i>P. falciparum</i> infection over 6 months (dose- and setting-dependent) [12]
Transmission-blocking vaccine	Pfs230D1-EPA (Alhydrogel)	Sexual-stage; mosquito midgut	Phase 1	Functional transmission-reducing activity (TRA): > 90% in 2/5 vaccinees (e.g., 99%, 98%) [13]
Transmission-blocking vaccine	Pfs25-EPA (Alhydrogel)	Sexual-stage; mosquito midgut	Phase 1	Lower functional activity than Pfs230D1-EPA in same study (0/5 > 50% TRA) [13]

Unlike small molecules and mAbs, malaria vaccine dose selection does not primarily rely on traditional PK-based or allometric scaling approaches. Instead, preclinical studies are used to evaluate immunogenicity, functional antibody responses, adjuvant selection, and safety, which guide clinical dose selection. For RTS,S/AS01, the antigen dose was established based on favorable immune responses and tolerability observed in preclinical and early clinical studies and was subsequently maintained across clinical populations. Consequently, optimization efforts focused on vaccination schedules and booster timing rather than antigen dose adjustment. A similar development strategy was used for R21/Matrix-M, where preclinical testing of multiple adjuvants identified Matrix-M as the optimal formulation based on low profile immunogenicity, with further Phase 2b studies (Table 1) demonstrating a favorable efficacy profile and strong immune responses, including in age youngest children cohorts aged (05–17) months [10].

Similarly to RTS,S/AS01, once R21/Matrix-M is administered into the body, systemic antigen concentrations are typically below the limit of detection. However, this does not prevent the induction of robust vaccine-induced immune responses, the magnitude and durability of which vary between individuals and across transmission settings. For vaccines, E-R relationships cannot be characterized using conventional PK metrics. Instead, they are inferred from vaccine-induced immune responses (Figure 3). These immunological readouts include antigen-specific IgA, IgM, and IgG titers, as well as functional antibody properties such as avidity and complement-fixing capacity, which collectively contribute to parasite neutralization and clearance [10, 14].

R21/Matrix-M induces broad humoral immunity, including IgG, IgM, and IgA responses, complement-fixing antibodies, and recognition of diverse malaria antigens. In contrast, RTS,S/AS01 primarily elicits CSP-specific and hepatitis B surface antigen-specific antibodies, together with CD4⁺ T-cell responses that have been associated with protection in children.

Both vaccines have demonstrated acceptable safety profiles and clinically meaningful efficacy in children. RTS,S has shown approximately 56% efficacy over 12 months in young children in Phase 3 trials, whereas R21/Matrix-M has demonstrated higher short-term efficacy, with reductions in symptomatic malaria of approximately 75% in seasonal transmission settings and 68% in perennial transmission settings in Phase 2b and Phase 3 studies [10, 15].

4.2 | mAbs—Pharmacokinetics, Pharmacodynamics, Exposure-Response, and Immunogenicity

Long-acting mAbs developed for malaria prevention exhibit well-characterized and quantifiable PK, in contrast to vaccines. Following intravenous (IV) or subcutaneous (SC) administration, long-acting mAbs such as CIS43LS and L9LS demonstrate predictable absorption, distribution, and elimination profiles consistent with IgG-based therapeutics [8, 12]. Fc engineering strategies, including LS mutations, enhance Fc receptor (FcRn)-mediated recycling, reduce lysosomal degradation, and

extend systemic half-life to several weeks or months [16, 17], allowing antibody concentrations to decline gradually in a dose-dependent manner. This predictable decay enables estimation of the duration over which protective serum concentrations are maintained, thereby supporting exposure–response modeling and rational dose selection [8, 12].

In clinical studies of CIS43LS, PK sampling was conducted before dosing and longitudinally up to 36 weeks post-administration, with antibody titers at early and later time points including day 7 and weeks 4 and 24. The PK of CIS43LS is linear with a $C_{\max}$ occurring at day 7 following a SC administration. Serum concentrations remain detectable for at least 24 weeks post-dose. Phase II analyses demonstrated dose-dependent protective efficacy, with estimated protection of 88.2% for the 40 mg/kg dose and 75% for the 10 mg/kg dose relative to placebo [8].

In clinical studies, L9LS has demonstrated dose-proportional (linear) PK across the evaluated dose range, with an estimated terminal half-life of approximately 56 days [12]. This extended half-life, conferred by Fc engineering to enhance neonatal Fc receptor (FcRn) recycling, supports sustained serum concentrations and makes L9LS particularly well suited for model-informed optimization of dosing strategies [11]. Such characteristics are aligned with seasonal malaria chemoprevention-like deployment paradigms, where a single pre-transmission administration could provide protection throughout a defined high-risk period.

Compared with CIS43LS, L9LS exhibits greater in vitro potency against *Plasmodium falciparum* sporozoites, with lower ED₅₀ and EC₅₀ values, reflecting enhanced neutralizing activity at lower concentrations [12, 18]. This increased potency translates into protective efficacy at reduced dose levels in clinical settings. Phase II studies evaluating subcutaneous administration have reported protective efficacy ranging from 66% to 77% [11] in pediatrics while Phase I studies in adults have reported efficacy of up to 88%, supporting the potential of this monoclonal antibody as a scalable preventive intervention [12]. Hence, the favorable PK profile, improved potency, and encouraging clinical efficacy position L9LS as a promising candidate for seasonal and potentially perennial malaria prevention strategies.

Collectively, these findings highlight fundamental differences between vaccines and monoclonal antibodies in malaria prevention. Vaccine-induced protection depends on complex and evolving immune responses, whereas mAb-mediated protection is directly linked to measurable antibody concentrations. Therefore, mAbs are particularly well suited for population PK/PD modeling, enabling quantitative characterization of exposure–response relationships, prediction of protection duration, and optimization of dose selection and deployment strategies. These features underscore the growing role of quantitative pharmacology in advancing antibody-based malaria prevention.

5 | Model-Informed Translational PK/PD Strategies for Anti-Malarial mAbs Development

A translational PK/PD framework has become an essential component of modern drug development, supporting both

mechanistic understanding and clinical decision-making from early human studies through later stages of development [19, 20]. The integration of PK and PD data at the individual and population levels enables robust MIDD strategy, including quantitative characterization of E-R relationships, estimation of the probability of protective threshold, and Monte Carlo-based clinical trial simulation to anticipate outcomes across diverse patient subgroups [20]. In the context of anti-malarial interventions, achieving and maintaining therapeutic concentrations for an adequate duration is pivotal for parasite clearance and prevention of breakthrough infection, and this is influenced by thoughtful selection of dose, dosing frequency, route of administration, and, when relevant, combination regimens [21]. As such, a rigorous understanding of PK/PD relationships accelerates rational design and optimization of regimens in early clinical development [9].

Population PK (Pop PK) models derived from FIH data can directly inform the design of CHMI trials by guiding model-informed selection of dose levels, routes of administration (e.g., IV versus SC), and timing of challenge inoculation to ensure informative exposure ranges and enhance interpretability of efficacy endpoints such as time to qPCR-detectable parasitemia [8, 22]. CHMI serves as a highly controlled proof-of-concept platform for monoclonal antibody research, enabling early quantification of exposure–protection relationships that can subsequently be translated to field settings with greater confidence [8].

For mAbs, PK is typically described using compartmental models that capture distribution between central and peripheral compartments and elimination processes. Pop PK model development is generally performed in a stepwise manner, beginning with simpler structural models, such as one- or two-compartment models with first-order absorption for subcutaneous administration and linear elimination. While linear first-order elimination often adequately describes disposition for many mAbs, nonlinear processes such as target-mediated drug disposition (TMDD) may be incorporated when supported by data, particularly at lower concentrations or in the presence of high target burden [19]. Integrated PK/PD models linked to appropriate PD structures (e.g., indirect response models) allow quantitative characterization of longitudinal data, facilitate exploration of dose- or E-R relationships through simulation, and support efficient selection of dose levels for subsequent studies [19]. Model-informed trial design integrated in such analyses can improve efficiency by reducing sample size requirements and enhancing the interpretability of outcomes [23].

Pop PK models can be used to characterize variability in antibody exposure and predict the duration for which serum concentrations remain above protective thresholds. These models are particularly valuable for evaluating subcutaneous absorption and bioavailability, as well as quantifying the impact of clinically relevant covariates such as body weight, age, and transmission intensity. Such simulations provide a quantitative framework for optimizing dose levels and dosing intervals to maintain protection throughout periods of malaria risk. Model performance is typically assessed using goodness-of-fit diagnostics, visual predictive checks, and comparisons between observed and model-predicted concentrations to ensure adequate characterization of antibody disposition and variability.

Systemic exposure to long-acting mAbs such as CIS43LS and L9LS is likely to vary across subpopulations, including children and individuals with altered physiology. Mechanistic PBPK models provide a complementary framework by integrating physiological parameters with drug-specific properties to mechanistically describe mAb disposition and elimination. These models can incorporate FcRn binding processes, evaluate key determinants of antibody persistence, and support extrapolation across age groups [16, 20].

Covariate effects such as age, body weight, and immune status play important roles in determining antimalarial mAb exposure. Pediatric patients often exhibit higher volume of distribution and altered clearance relative to adults because of developmental changes in physiological processes, including FcRn-mediated recycling, the pathway responsible for rescuing endogenous IgG and therapeutic antibodies from intracellular degradation. Consequently, simple body weight-based scaling may be insufficient to accurately predict exposure in children. Immune status may also influence antibody disposition, as malaria infection and immune activation can increase endogenous IgG production, potentially altering FcRn-mediated recycling and antibody persistence. Pop PK and nonlinear mixed-effects modeling approaches can be used to quantify these sources of variability and identify clinically relevant covariates affecting exposure [16, 17, 20].

PBPK approaches have demonstrated increasing utility in predicting age-dependent PK and informing pediatric dose selection, particularly when clinical data are limited. In the context of malaria prevention, PBPK models can be used to predict doses required to achieve and maintain protective anti-CSP antibody concentrations during periods of high transmission risk. By linking predicted exposure profiles to protective thresholds, these models may also support estimation of protection duration and optimization of redosing intervals and sustain prophylactic coverage throughout the malaria transmission season.

Route of administration may also differentially affect exposure in children and special populations. For subcutaneous or intramuscular delivery, lymphatic uptake precedes systemic availability, and developmental differences in lymphatic transport can alter absorption kinetics and peak exposure [19, 24]. Taken together, these considerations highlight the importance of PBPK and population PK/PD modeling to characterize variability, predict therapeutic exposure, and support rational dose selection for monoclonal antibody-based malaria prevention across pediatric and other special populations.

6 | Forward-Looking Perspectives on Vaccines and mAbs in Malaria Prevention

Vaccines and long-acting mAbs represent two mechanistically distinct but potentially complementary strategies in the fight against malaria (Figure 2). Pre-erythrocytic vaccines such as RTS,S/AS01 and R21/Matrix-M induce active immunity by stimulating polyclonal antibody responses and CD4⁺ T-cell activity against the CSP, thereby reducing the risk of infection and clinical disease at the population level [7, 10]. In contrast, long-acting mAbs such as CIS43LS and L9LS provide immediate, passive immunity through direct neutralization of *Plasmodium*

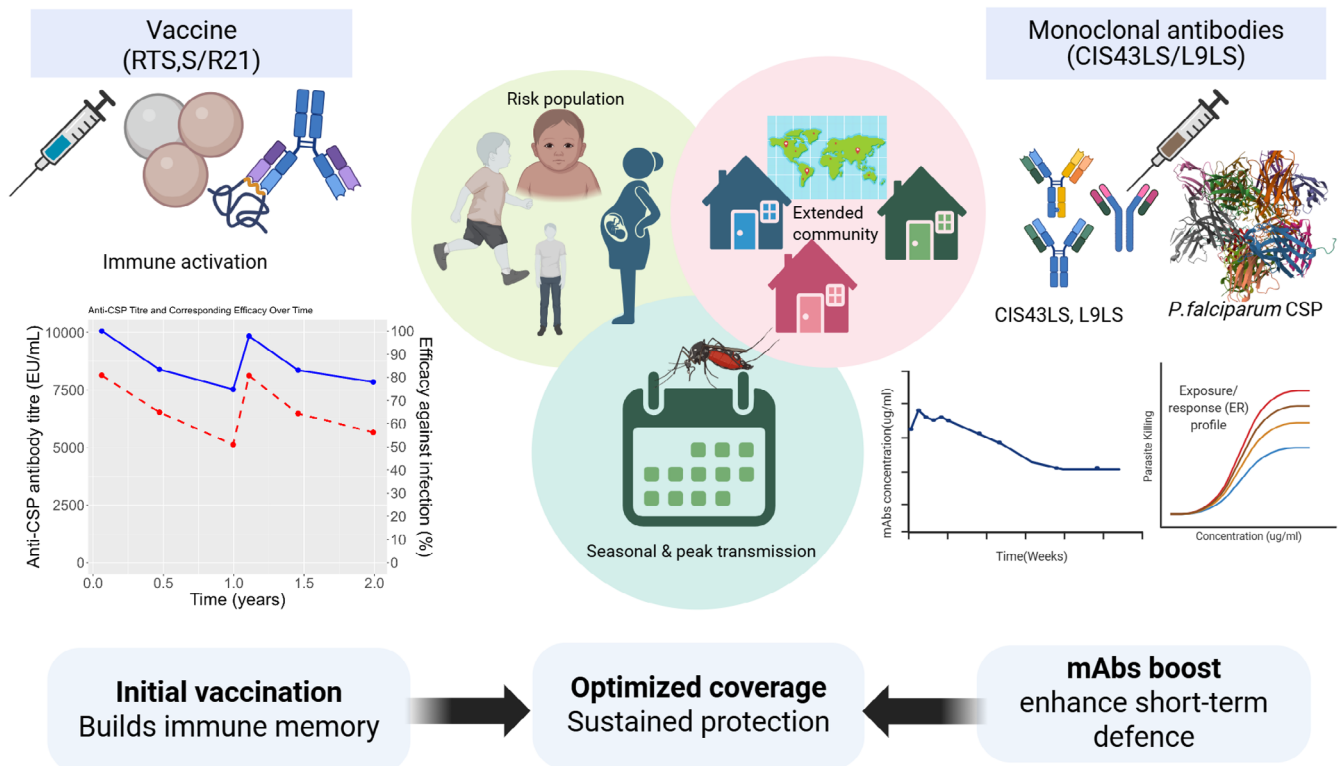


FIGURE 2 | Complementary combination of malaria vaccines and monoclonal antibodies. Vaccines (left) induce active immunity over time, with gradual development of antigen-specific antibodies and T-cell responses, leading to durable but delayed protection. Monoclonal antibodies (right) provide immediate, passive protection with high systemic concentrations shortly after administration, offering rapid but time-limited protection. A combination strategy illustrates how monoclonal antibodies can bridge the period before vaccine-induced immunity is established, providing continuous protection during high-risk transmission periods.

falciparum sporozoites, with protection that is quantitatively linked to systemic exposure and duration above protective thresholds [8, 16].

While vaccines offer scalability and the potential for durable immune memory, mAbs provide predictable, exposure-driven protection that is especially valuable for high-risk populations such as infants, immunocompromised individuals, or travelers, and during seasonal transmission peaks. These modalities can therefore be deployed synergistically: vaccines establish broad population-level immunity, while long-acting mAbs bridge gaps in protection or provide rapid prophylaxis when immediate protection is required. Combining these approaches may enhance overall effectiveness, particularly in areas where vaccine efficacy is moderate, wanes over time, or where rapid immunity is needed before peak transmission periods [22].

From a translational pharmacology perspective, the development of long-acting mAbs has highlighted the importance of quantitative PK/PD modeling. Pop PK and PBPK models allow prediction of exposure across diverse populations, including pediatric, pregnant, and immunologically naive individuals, and inform rational dose selection, route of administration, and timing of administration relative to transmission seasonality [20, 24–25]. Integration of E-R data from CHMI trials further enables early proof-of-concept evaluation and model-informed optimization of regimens before large-scale field trials [22, 24].

Building upon these principles, Model-Informed Vaccine and Monoclonal Antibody Development (MIVD) provides a quantitative framework for evaluating combined vaccine–mAb prevention strategies. MIVD leverages exposure–response principles to balance vaccine immunogenicity and reactogenicity, as illustrated in Figure 3, while characterizing exposure-driven protection for mAbs. Mathematical and systems-based models can integrate vaccine-induced antibody kinetics, including peak CSP antibody titers and decay rates, with mAb pharmacokinetics to define protective thresholds and predict the duration of protection. Within a hypothetical seasonal prophylaxis scenario, quantitative systems pharmacology (QSP) models can evaluate additive or synergistic effects of vaccines and mAbs, characterize how complementary administration influences protection during periods of high malaria transmission, and predict the probability of sustaining prophylactic coverage throughout a transmission season. Virtual trial simulations can further support optimization of deployment strategies, booster timing, and dosing intervals, thereby strengthening the application of MIDD principles to the development of next-generation malaria prevention programs.

7 | Future Directions

Looking forward, several areas hold promises for advancing malaria prevention:

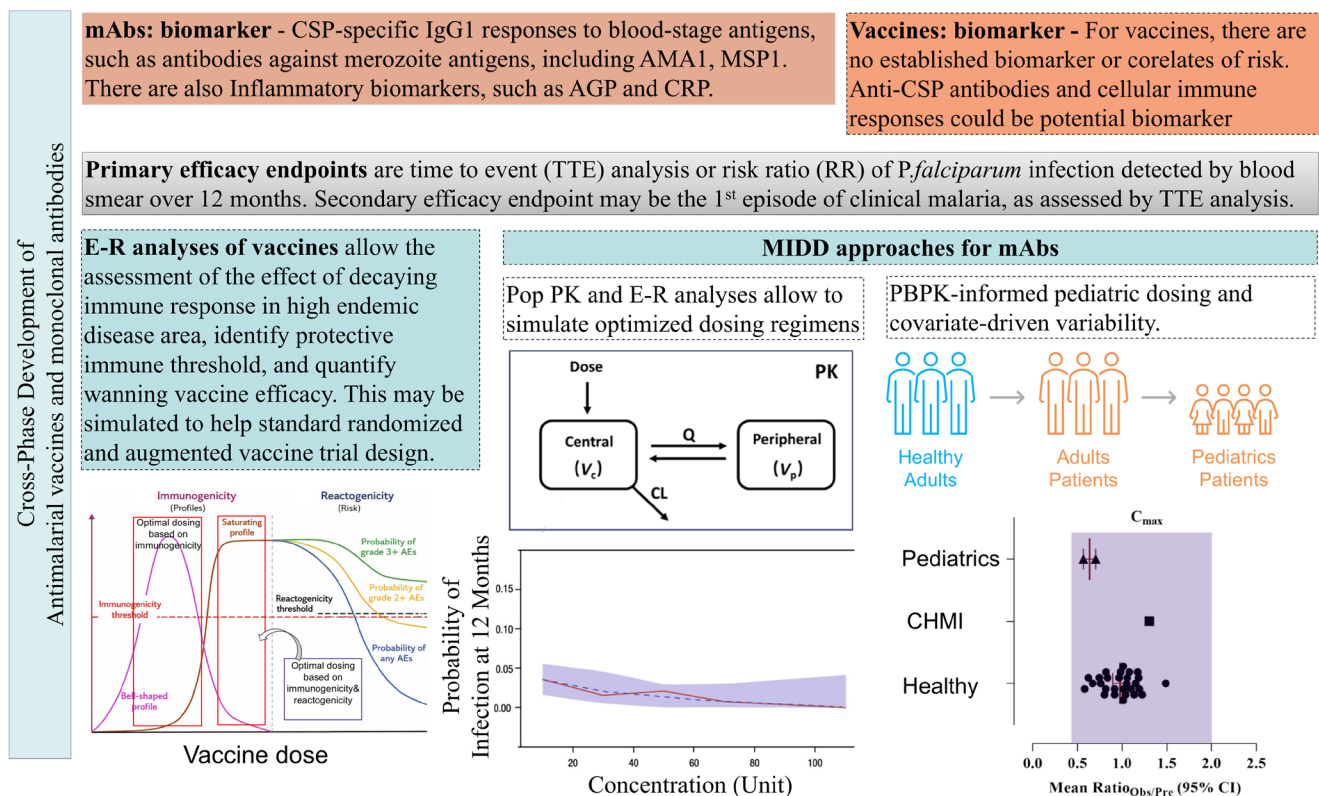


FIGURE 3 | Overview of quantitative and model-informed approaches for malaria vaccines and monoclonal antibodies. The figure highlights biomarkers, efficacy endpoints, exposure–response analyses, and the application of population pharmacokinetic (Pop PK) and physiologically based pharmacokinetic (PBPK) models to support dose optimization, pediatric extrapolation, and sustained prophylactic coverage. AEs, adverse events; AGP, alpha-1-acid glycoprotein; AMA1, apical membrane antigen 1; CHMI, controlled human malaria infection; CI, confidence interval; CL, clearance; C_{max} , maximum concentration; CRP, C-reactive protein; IgG1, immunoglobulin G subclass 1; mAbs, monoclonal antibodies; MIDD, model-informed drug development; MSP1, merozoite surface protein 1; Obs/Pre, observed-to-predicted ratio; Q, intercompartmental clearance; V_c , central volume of distribution; V_p , peripheral volume of distribution.

- Combination Therapy:** Strategic co-administration of vaccines and mAbs could leverage complementary mechanisms, achieving both immediate and long-term protection. Optimization of sequential or simultaneous use could be guided by mechanistic PK/PD and immuno-epidemiologic models.
- Next-Generation Antibodies:** Engineering of mAbs for extended half-life, improved potency, or multi-epitope targeting could enhance durability and breadth of protection, reducing the need for frequent administration.
- Personalized or Population-Tailored Strategies:** PBPK modeling and exposure simulations can support dose adjustments in pediatrics, pregnancy, and immunocompromised populations to ensure adequate protection, bridging the gap between adult clinical data and special populations.
- Integration with Epidemiologic Models:** Coupling PK/PD-informed protection estimates with population-level transmission models can optimize public health deployment strategies, including seasonal or geographically targeted interventions.
- Monitoring and Correlates of Protection:** Continued efforts to define immune correlates for both vaccines and

mAbs will enhance predictive modeling and accelerate regulatory approval pathways.

8 | Conclusion

The development of vaccines and long-acting mAbs illustrates two distinct PD paradigms in malaria prevention: vaccine efficacy is driven by complex, time-varying immune responses, whereas mAb-mediated protection is directly tied to measurable systemic exposure. Rather than competing modalities, these interventions are complementary, offering opportunities for integrated strategies that maximize population coverage and individual protection. Translational PK/PD and PBPK modeling will continue to play a central role in optimizing dosing, predicting protection in diverse populations, and informing clinical trial and public health strategies. Future innovation, including next-generation antibodies and combination therapy with vaccines, has the potential to substantially accelerate progress toward malaria elimination.

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Conflicts of Interest

S.A.-O. is an employee of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co. Inc., Rahway, NJ, USA and may hold stock options in Merck & Co. Inc., Rahway, NJ, USA. All other authors declared no competing interests for this work.

References

1. Monoclonal antibody prevents malaria infection in African adults | National Institutes of Health (NIH), (2022), <https://www.nih.gov/news-events/news-releases/monoclonal-antibody-prevents-malaria-infection-african-adults>.
2. World Health Organization, “Malaria,” (2024), <https://www.who.int/news-room/fact-sheets/detail/malaria>.
3. A. J. Birkett, E. Ansah, S. Gordon, et al., “Value Profile for Malaria Vaccines and Monoclonal Antibodies,” *Vaccine* 3 (2025): 127971, <https://doi.org/10.1016/j.vaccine.2025.127971>.
4. K. Miura, Y. Flores-Garcia, C. A. Long, and F. Zavala, “Vaccines and Monoclonal Antibodies: New Tools for Malaria Control,” *Clinical Microbiology Reviews* 37 (2024): e0007123.
5. M. B. Laurens, “The Promise of a Malaria Vaccine-Are we Closer?,” *Annual Review of Microbiology* 72 (2018): 273–292.
6. RTS,S Clinical Trials Partnership, “Efficacy and Safety of the RTS,S/AS01 Malaria Vaccine During 18 Months After Vaccination: A Phase 3 Randomized, Controlled Trial in Children and Young Infants at 11 African Sites,” *PLoS Medicine* 11 (2014): e1001685.
7. RTS,S Clinical Trials Partnership, “Efficacy and Safety of RTS,S/AS01 Malaria Vaccine With or Without a Booster Dose in Infants and Children in Africa: Final Results of a Phase 3, Individually Randomised, Controlled Trial,” *Lancet* 386 (2015): 31–45.
8. M. R. Gaudinski, N. M. Berkowitz, A. H. Idris, et al., “A Monoclonal Antibody for Malaria Prevention,” *New England Journal of Medicine* 385 (2021): 803–814.
9. E. Hughes, E. Wallender, A. Mohamed Ali, P. Jagannathan, and R. M. Savic, “Malaria PK/PD and the Role Pharmacometrics Can Play in the Global Health Arena: Malaria Treatment Regimens for Vulnerable Populations,” *Clinical Pharmacology and Therapeutics* 110 (2021): 926–940.
10. M. S. Dattoo, M. H. Natama, A. Somé, et al., “Efficacy of a Low-Dose Candidate Malaria Vaccine, R21 in Adjuvant Matrix-M, With Seasonal Administration to Children in Burkina Faso: A Randomised Controlled Trial,” *Lancet* 397 (2021): 1809–1818.
11. K. Kayentao, A. Ongoiba, A. C. Preston, et al., “Subcutaneous Administration of a Monoclonal Antibody to Prevent Malaria,” *New England Journal of Medicine* 390 (2024): 1549–1559.
12. R. L. Wu, A. H. Idris, N. M. Berkowitz, et al., “Low-Dose Subcutaneous or Intravenous Monoclonal Antibody to Prevent Malaria,” *New England Journal of Medicine* 387 (2022): 397–407.
13. National Institute of Allergy and Infectious Diseases (NIAID), “Phase 1 Dose Escalating, Double-Blind, Randomized Comparator Controlled Trial of the Safety and Immunogenicity of Pfs25M-EPA/AS01 and Pfs230D1M-EPA/AS01 Vaccines, Transmission Blocking Vaccines Against Plasmodium Falciparum at Full and Fractional Dosing in Adults in Mali,” (2024), <https://clinicaltrials.gov/study/NCT02942277>.
14. M. T. White, P. Bejon, A. Olotu, et al., “The Relationship Between RTS,S Vaccine-Induced Antibodies, CD4⁺ T Cell Responses and Protection Against *Plasmodium falciparum* Infection,” *PLoS One* 8 (2013): e61395.
15. M. S. Dattoo, A. Dicko, H. Tinto, et al., “Safety and Efficacy of Malaria Vaccine Candidate R21/Matrix-M in African Children: A Multi-centre, Double-Blind, Randomised, Phase 3 Trial,” *Lancet* 403 (2024): 533–544.
16. G. J. Robbie, R. Criste, W. F. Dall’Acqua, et al., “A Novel Investigational Fc-Modified Humanized Monoclonal Antibody, Motavizumab-YTE, Has an Extended Half-Life in Healthy Adults,” *Antimicrobial Agents and Chemotherapy* 57 (2013): 6147–6153.
17. S.-Y. Ko, A. Pegu, R. S. Rudicell, et al., “Enhanced Neonatal Fc Receptor Function Improves Protection Against Primate SHIV Infection,” *Nature* 514 (2014): 642–645.
18. J. Skinner, K. Kayentao, A. Ongoiba, et al., “Anti-Sporozoite Monoclonal Antibody for Malaria Prevention: Secondary Efficacy Outcome of a Phase 2 Randomized Trial,” *Nature Medicine* 31 (2025): 2682–2690.
19. B. Pasquiers, S. Benamara, M. Felices, L. Nguyen, and X. Declèves, “Review of the Existing Translational Pharmacokinetics Modeling Approaches Specific to Monoclonal Antibodies (mAbs) to Support the First-In-Human (FIH) Dose Selection,” *International Journal of Molecular Sciences* 23 (2022): 12754.
20. Z. H. Temrikar, S. Suryawanshi, and B. Meibohm, “Pharmacokinetics and Clinical Pharmacology of Monoclonal Antibodies in Pediatric Patients,” *Paediatric Drugs* 22 (2020): 199–216.
21. N. J. White, “Pharmacokinetic and Pharmacodynamic Considerations in Antimalarial Dose Optimization,” *Antimicrobial Agents and Chemotherapy* 57 (2013): 5792–5807.
22. X. Tian, H. E. Janes, and J. G. Kublin, “Statistical Design and Analysis of Controlled Human Malaria Infection Trials,” *Malaria Journal* 23 (2024): 133.
23. A. Lim, P. Sharma, O. Stepanov, and V. P. Reddy, “Application of Modelling and Simulation Approaches to Predict Pharmacokinetics of Therapeutic Monoclonal Antibodies in Pediatric Population,” *Pharmaceutics* 15 (2023): 1552.
24. M. Kumar, S. Lanke, and D. K. Shah, “Investigation of Monoclonal Antibody Pharmacokinetics in Pediatric Population and Characterization Using a Platform PBPK Model,” *AAPS Journal* 28 (2026): 44.
25. C. Zunino, V. Gualano, H. Zhou, V. Lukacova, and M. Le Merdy, “Prediction of Monoclonal Antibody Pharmacokinetics in Pediatric Populations Using PBPK Modeling and Simulation,” *Pharmaceutics* 17 (2025): 884.